Supplementary material: Retention-constrained class expansion with reference-anchored candidate tables

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Sections A–H retain the original three-partition ViT-L/14 auxiliary and validation studies. Section I reports the expanded ten-partition, two-encoder comparison; Section J adds the disjoint-pool risk-calibration study. Their statistical units are stated separately.

Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 4.5. All experiments share the same frozen representations, true class partitions and disjoint support pools.

Appendix A.1. Readouts and evaluation measures

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced k-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the base classifier and calibration protocol with reference-anchored fusion. This comparison changes the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy A_B, macro novel accuracy A_N, all-class macro accuracy, and the harmonic mean in Eq. (14). Both datasets use equal numbers of base and novel classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

Appendix A.2. Recognition after five-shot class expansion

Reference-anchored fusion reaches 86.03 ± 1.40 base accuracy and 66.90 ± 4.23 novel accuracy (Table S1). Its base accuracy is 7.71 percentage points

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

Readout	Base	Novel	All	Harmonic
CLIP text	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Visual prototypes	77.53 ± 2.03	60.92 ± 3.94	69.23 ± 2.09	68.17 ± 2.49
Text + prototypes	82.47 ± 0.87	77.53 ± 0.47	80.00 ± 0.56	79.92 ± 0.55
Tip-Adapter (tuned)	78.38 ± 0.24	77.42 ± 1.72	77.90 ± 0.83	77.89 ± 0.83
Raw evidence fusion	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Candidate-normalized table	85.36 ± 0.66	70.83 ± 4.33	78.09 ± 2.02	77.36 ± 2.48
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches 80.00 ± 0.56 all-class accuracy and 77.53 ± 0.47 novel accuracy. Candidate-normalized fusion reaches 85.36 ± 0.66 base and 70.83 ± 4.33 novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets

On fine-grained Pets expansion, reference anchoring reaches 97.18 ± 2.25 base and 87.79 ± 6.18 novel accuracy (Table S2). Base accuracy is 2.41 points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

Readout	Base	Novel	All	Harmonic
CLIP text	92.87 ± 3.37	94.05 ± 1.35	93.46 ± 1.59	93.43 ± 1.63
Visual prototypes	92.08 ± 2.61	68.30 ± 4.62	80.19 ± 2.33	78.35 ± 2.97
Text + prototypes	94.77 ± 2.91	93.91 ± 1.55	94.34 ± 1.43	94.31 ± 1.44
Tip-Adapter (tuned)	93.56 ± 2.45	94.07 ± 1.18	93.81 ± 1.27	93.80 ± 1.28
Raw evidence fusion	90.13 ± 8.01	91.18 ± 4.27	90.66 ± 6.13	90.63 ± 6.18
Candidate-normalized table	96.85 ± 1.50	86.49 ± 7.64	91.67 ± 4.56	91.29 ± 4.99
Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	92.14 ± 3.23

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

Readout	Base	Novel	All
Text + group offset	78.93 ± 2.75	75.56 ± 4.40	77.24 ± 0.89
Anchored text + prototypes	82.40 ± 2.02	76.19 ± 3.18	79.30 ± 0.59
Frozen RP-ridge (2000)	70.18 ± 0.69	52.52 ± 4.27	61.35 ± 1.83

Oxford-IIIT Pets

Readout	Base	Novel	All
Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31
Anchored text + prototypes	94.58 ± 3.22	90.38 ± 5.41	92.48 ± 1.10
Frozen RP-ridge (2000)	93.49 ± 3.32	73.70 ± 5.13	83.60 ± 1.13

Text–prototype fusion reaches 94.34 ± 1.43 all-class macro accuracy; the reference-anchored table reaches 92.49 ± 2.83 with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only calibration rule.

Appendix A.4. Calibration controls and analytic incremental readouts

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and visual evidence for every class. Harmonic selection searches offsets from -4 to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

Data	Readout	Base	Novel	All	Feasible
CIFAR	Text + group offset	81.47 ± 1.13	71.49 ± 5.03	76.48 ± 1.99	1/3
CIFAR	Anchored text + prototypes	82.84 ± 1.50	77.11 ± 4.46	79.98 ± 1.61	3/3
CIFAR	Candidate-normalized table	84.69 ± 1.92	72.45 ± 4.14	78.57 ± 1.54	3/3
CIFAR	Reference-anchored table	84.56 ± 1.31	73.88 ± 5.01	79.22 ± 2.21	3/3
Pets	Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31	3/3
Pets	Anchored text + prototypes	94.03 ± 3.37	90.96 ± 5.91	92.50 ± 1.27	3/3
Pets	Candidate-normalized table	97.25 ± 1.78	85.26 ± 7.25	91.26 ± 4.40	3/3
Pets	Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	3/3

across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and 95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode selects a novel offset. This frozen adaptation uses base validation in place of the original within-task MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

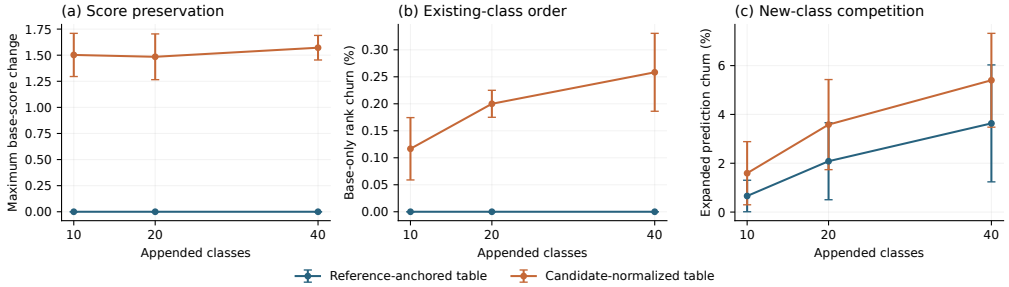


Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99% results accompany the experimental records.

Appendix A.5. Vocabulary growth and score preservation

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

Reference anchoring gives zero measured base-score drift and zero base-only ranking churn across all nine CIFAR partition and vocabulary settings (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

Shots	Base	Novel	All	Harmonic
1	84.57 ± 0.88	70.86 ± 5.41	77.72 ± 2.27	77.00 ± 2.85
2	85.65 ± 1.45	70.05 ± 1.13	77.85 ± 0.67	77.04 ± 0.67
5	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of 2.85×10^{-5} on CIFAR and 3.58×10^{-6} on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

Appendix A.6. Support budget and evidence availability

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero branch coefficients, allowing the selected mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

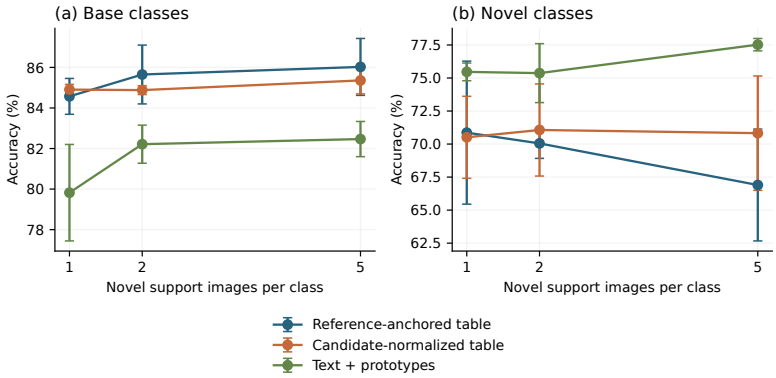


Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

Readout	Base	Novel	All	Harmonic
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14
Without learned evidence	58.10 ± 32.21	80.72 ± 4.47	69.41 ± 13.87	62.85 ± 25.08
Without novel prototypes	86.42 ± 1.71	47.17 ± 30.85	66.79 ± 14.62	56.15 ± 30.05
Without offset	86.14 ± 1.69	65.58 ± 7.35	75.86 ± 2.83	74.25 ± 4.08

Appendix A.7. Mechanism ablation and operating trade-off

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

Figure S3 sweeps d from -2 to 2 in steps of 0.5 with a and u fixed. Increasing d shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (13)). The resulting curve identifies

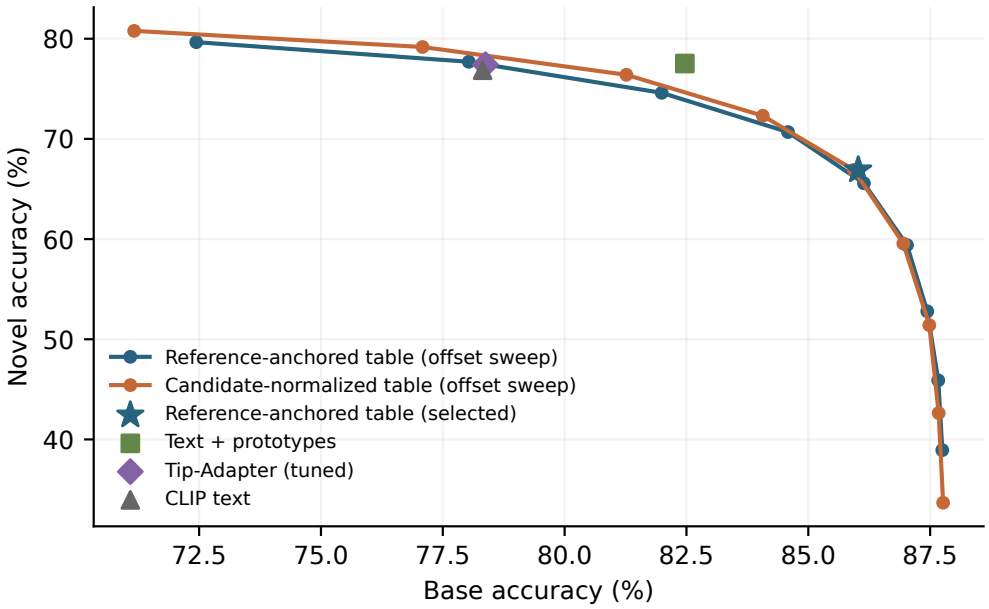


Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

how the offset controls admission of new categories; a validation retention target selects the operating point.

Appendix A.8. Readout efficiency

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization. They include reference statistics and fusion and exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR	90%	Retained classifier	78.33 ± 3.89	80.13 ± 2.25	3/3
CIFAR	90%	Classifier removed (fixed gates)	79.07 ± 3.94	70.79 ± 13.81	3/3
CIFAR	90%	No classifier (recalibrated)	78.73 ± 3.21	75.45 ± 5.43	3/3
CIFAR	90%	Anchored text + prototypes	80.18 ± 0.60	79.45 ± 0.76	3/3
CIFAR	95%	Retained classifier	82.99 ± 2.29	75.71 ± 3.14	3/3
CIFAR	95%	Classifier removed (fixed gates)	81.95 ± 0.04	71.54 ± 3.87	2/3
CIFAR	95%	No classifier (recalibrated)	81.85 ± 0.18	71.70 ± 4.10	2/3
CIFAR	95%	Anchored text + prototypes	81.95 ± 0.64	77.48 ± 1.52	2/3
CIFAR	99%	Retained classifier	86.48 ± 1.65	62.83 ± 10.11	3/3
CIFAR	99%	Classifier removed (fixed gates)	–	–	0/3
CIFAR	99%	No classifier (recalibrated)	–	–	0/3
CIFAR	99%	Anchored text + prototypes	86.85	53.85	1/3
Pets	90%	Retained classifier	84.87 ± 4.59	96.53 ± 0.78	3/3
Pets	90%	Classifier removed (fixed gates)	84.11 ± 2.76	96.44 ± 0.64	3/3
Pets	90%	No classifier (recalibrated)	82.00 ± 0.20	96.38 ± 0.52	3/3
Pets	90%	Anchored text + prototypes	85.69 ± 4.62	95.87 ± 1.57	3/3
Pets	95%	Retained classifier	90.29 ± 2.90	95.68 ± 0.83	3/3
Pets	95%	Classifier removed (fixed gates)	90.11 ± 1.17	95.71 ± 0.11	2/3
Pets	95%	No classifier (recalibrated)	89.88 ± 0.85	95.71 ± 0.11	2/3
Pets	95%	Anchored text + prototypes	89.37 ± 4.15	95.63 ± 1.55	3/3
Pets	99%	Retained classifier	96.43 ± 2.15	88.67 ± 7.89	3/3
Pets	99%	Classifier removed (fixed gates)	–	–	0/3
Pets	99%	No classifier (recalibrated)	–	–	0/3
Pets	99%	Anchored text + prototypes	97.70 ± 1.11	84.19 ± 12.46	2/3

Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are “a photo of a {}.”, “a bad photo of a {}.”, “a cropped photo of a {}.”, “a close-up photo of a {}.”, “a bright photo of a {}.”, “a good photo of a {}.”, “a photo of the {}.”, and a second occurrence of “a

photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential prototype products can exhibit the reported floating-point residuals when matrix widths change.

Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing d cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase M_N but cannot change M_B. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

Appendix D.1. Exact empirical retention envelope

For a fixed readout, let b_i and n_i be the canonical winning base and novel class for query i, and let $\Delta_i = M_B(x_i) - M_N(x_i)$. Define $g_i(d) = \mathbf{1}[d > \Delta_i \text{ or } (d = \Delta_i \text{ and } n_i < b_i)]$. With the class-macro weights w_i, the two group accuracies are

$$A_B(d) = \sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i](1 - g_i(d)), \tag{D.1}$$

$$A_N(d) = \sum_{i:y_i \in \mathcal{N}} w_i \mathbf{1}[n_i = y_i] g_i(d). \tag{D.2}$$

Thus A_B is nonincreasing and A_N is nondecreasing in d. For the stored finite-precision scores, an exact search need only consider each observed margin and its immediately preceding representable offset, together with the all-base and all-novel endpoints. Between adjacent margins every group decision is fixed. If τ exceeds $\sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i]$, the constraint is infeasible. Otherwise, sorting the margins of correctly classified base queries in descending order and accumulating their macro weights locates the boundary at which retained base mass first reaches τ. The tie rule selects that boundary or its predecessor. This is the largest feasible offset in the implementation's offset ordering. Monotonicity of A_N proves that it maximizes novel accuracy under the empirical constraint. The argument applies identically to base-validation selection; that policy's test outcome is evaluated on separate queries.

The envelope separates the within-base feasible ceiling, the admitted group threshold and the distribution of correctly classified novel-query margins. It therefore measures the joint operating value of a learned readout and its competition with arriving evidence, beyond the statement that stored scores are preserved.

Appendix E. Additional background and score-preservation proof

Appendix E.1. Long-tailed recognition and calibration

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pretrained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

Appendix E.2. Semantic and visual evidence for few-shot recognition

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt

context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

Appendix E.3. Proof of the main score-preservation proposition

Proof. For an existing candidate c, each raw score $q_e(x, c)$ and mask m_{ec} is fixed. The means and variances in Eq. (9) are functions of x and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (10) is unchanged. The coefficients and offset of c are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

Corollary 1. *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

Proof. The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

Appendix F. Recent readout comparison and ImageNet-R extension

The extension protocol was recorded before ImageNet-R evaluation. Class partitions use seeds 0, 1 and 2, with exact frequency-stratified allocation of 100 base and 100 novel classes. Within each base group, seed $900 + p$ assigns 50 pseudo-base and 50 pseudo-novel classes. Base-only validation selects ridge regularization, branch mixtures and readout hyperparameters. No ImageNet-R test result selects these quantities. The retained table, removal controls and dense prototype replacement use the same grids as the CIFAR/Pets experiment.

FeNeC [19] is independently implemented from the paper's equations and checked against author-code commit `1575ea019fd583149c47c068f7f5ccc102ea4580`. Both variants use identity Tukey transformation for signed CLIP features and Euclidean k-means with a maximum of 100 iterations. The representative grid is $K \in \{1, 5, 25\}$ and the neighbor grid is $\{1, 3, 5\}$. FeNeC-M estimates one covariance per class, applies the author's ViT single-shrinkage and correlation normalization with $(\gamma_1, \gamma_2) \in \{(1, 1), (6.12, 8.10), (9.98, 0)\}$, then uses inverse squared-Mahalanobis-distance votes. FeNeC-E uses inverse unsquared-Euclidean-distance votes. Their native predictions are preserved when votes are converted to log votes, with a floor of 10^{-30}. The calibrated forms select a novel offset from $[-4, 4]$ in steps of 0.25 on the pseudo episode. Harmonic macro accuracy chooses the configuration.

The five-shot protocol caps a class's representative count at its sample count. A class with one fitting sample uses identity covariance, avoiding an undefined sample covariance. These are explicit sparse-data adaptations. A per-class independent summary can be cached before its arrival; the voting neighborhood always contains only currently admitted classes. Native and calibrated forms share each current-vocabulary score matrix, so their complete matched-retention frontiers coincide, while fixed-offset trajectories can differ. The final grids are documented with the code. FeNeC frontiers use its sparse global-neighbor votes, with unvoted classes at the common floor; feasibility refers to this output interface.

All ImageNet-R images are content-hashed before pool assignment. Fifty-nine byte-identical duplicates are removed globally. No retained content hash appears in two fitting, validation, support or test pools. The source archive has SHA-256 `18c6bf493b39a0d975d48e587437f562caab9c52ae6327dcfa9dd8eb54aa1b52`. The class-name mapping, image paths, achieved fitting counts, source hashes, calibrated settings and prediction margins accompany the code and records. These are task-specific splits of ImageNet-R, whose source release is a rendition benchmark [20].

Table S8: ImageNet-R matched-retention recognition at three budgets after five arrivals. Values are percentages, mean ± partition-level sample standard deviation over feasible partitions. Counts specify available partition sets.

Budget	Readout	Base	Novel	Feasible
90%	Retained classifier	81.83 ± 0.29	86.80 ± 0.27	3/3
90%	Classifier removed (fixed gates)	81.81 ± 0.31	85.53 ± 0.49	3/3
90%	No classifier (recalibrated)	81.81 ± 0.31	85.34 ± 0.46	3/3
90%	Anchored text + prototypes	81.83 ± 0.32	86.59 ± 0.58	3/3
90%	Tip-Adapter sum + offset	81.82 ± 0.30	83.74 ± 0.25	3/3
90%	Tip-Adapter mean + offset	81.83 ± 0.30	85.30 ± 0.23	3/3
95%	Retained classifier	86.38 ± 0.32	84.48 ± 0.72	3/3
95%	Classifier removed (fixed gates)	86.37 ± 0.31	72.62 ± 12.51	3/3
95%	No classifier (recalibrated)	86.37 ± 0.31	72.69 ± 12.57	3/3
95%	Anchored text + prototypes	86.37 ± 0.32	83.32 ± 1.40	3/3
95%	Tip-Adapter sum + offset	86.41 ± 0.30	70.93 ± 10.32	3/3
95%	Tip-Adapter mean + offset	86.38 ± 0.33	81.58 ± 1.72	3/3
99%	Retained classifier	90.02 ± 0.34	73.69 ± 1.50	3/3
99%	Classifier removed (fixed gates)	–	–	0/3
99%	No classifier (recalibrated)	–	–	0/3
99%	Anchored text + prototypes	89.98 ± 0.47	47.62 ± 13.36	2/3
99%	Tip-Adapter sum + offset	–	–	0/3
99%	Tip-Adapter mean + offset	–	–	0/3

Table S9: ImageNet-R test outcomes with the 95% base-validation-only threshold policy. Validation feasibility is reported separately from achieved test accuracy. Tip-Adapter uses its base-only selected calibrated cache settings, followed by validation-only threshold selection.

Readout	Base test	Novel test	Validation feasible
Retained classifier	86.01 ± 0.59	84.73 ± 0.71	3/3
Classifier removed (fixed gates)	86.38 ± 0.18	57.35 ± 38.51	3/3
No classifier (recalibrated)	86.48 ± 0.20	57.10 ± 38.30	3/3
Anchored text + prototypes	86.36 ± 0.47	83.26 ± 1.49	3/3
Tip-Adapter sum + offset	86.10 ± 0.46	74.08 ± 5.48	3/3
Tip-Adapter mean + offset	86.68 ± 0.23	80.75 ± 2.09	3/3

Table S10: FeNeC matched-test-retention operating points at strict budgets. Values are percentages, mean ± sample standard deviation over feasible partitions; missing values have zero feasible partitions. Native and offset-calibrated forms share a score matrix and hence the same complete offset-sweep frontier.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR-100	95%	FeNeC-M + offset	–	–	0/3
CIFAR-100	95%	FeNeC-E + offset	–	–	0/3
CIFAR-100	99%	FeNeC-M + offset	–	–	0/3
CIFAR-100	99%	FeNeC-E + offset	–	–	0/3
Pets	95%	FeNeC-M + offset	–	–	0/3
Pets	95%	FeNeC-E + offset	94.23	69.46	1/3
Pets	99%	FeNeC-M + offset	–	–	0/3
Pets	99%	FeNeC-E + offset	–	–	0/3
ImageNet-R	95%	FeNeC-M + offset	–	–	0/3
ImageNet-R	95%	FeNeC-E + offset	–	–	0/3
ImageNet-R	99%	FeNeC-M + offset	–	–	0/3
ImageNet-R	99%	FeNeC-E + offset	–	–	0/3

Appendix G. Dense caches, repeated calibration and measured readout cost

Appendix G.1. Dense semantic-cache protocol

The training-free Tip-Adapter score follows the ECCV paper and the author implementation [17]: cosine similarity to labeled cache keys is transformed by $\exp[-\beta(1 - \text{similarity})]$, summed into classes and added to 100 times text similarity with weight α. The author code is pinned at commit `d0e2d6f8c5feb8b6ce937b757810761f7155d4d5`. All permitted base fitting images and five reserved novel support images supply the keys. The sum variant retains the published aggregation. A separate class-mean variant divides each class's retrieval score by its key count, isolating the effect of unequal long-tailed cache sizes. Both reuse the same single-view frozen features and text anchors as the tables. This comparison transfers the published readout under controlled feature access; the original augmented-cache extraction and trainable Tip-Adapter-F are distinct settings.

On each original base-only pseudo episode, we search $\beta \in \{0.1, 1, 2, 5, 10, 20, 50\}$ and $\alpha \in \{0.1, 1, 5, 10, 20, 50, 100, 200\}$. The native form selects both parameters with zero offset. The calibrated form jointly selects these parameters and an offset from $[-40, 40]$ in steps of 2.5, using macro harmonic accuracy. This 33-point offset grid reflects the raw 100-scaled CLIP score system. No actual novel query or test label is used in selection. Both forms are reported because their selected cache settings can differ. All matched-retention targets are the original retained readout's partition-specific absolute targets.

Table S11: Complete five-shot fixed-gate readout comparison. Values are percentages, mean ± sample standard deviation over three class partitions after averaging the two arrival orders. The paired gain is retained minus comparator average incremental accuracy; a negative gain favors the comparator. Native and calibrated cache forms are selected separately.

Data	Readout	Incremental avg.	Final	Paired avg. gain
CIFAR-100	Retained classifier	82.02 ± 0.84	79.31 ± 1.14	0.00 ± 0.00
CIFAR-100	Classifier removed (fixed gates)	71.40 ± 14.78	68.03 ± 15.63	10.62 ± 15.22
CIFAR-100	No classifier (recalibrated)	79.84 ± 0.82	77.56 ± 1.02	2.18 ± 0.24
CIFAR-100	Anchored text + prototypes	81.56 ± 0.52	79.30 ± 0.59	0.46 ± 0.44
CIFAR-100	Tip-Adapter sum + offset	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean + offset	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
CIFAR-100	FeNeC-M + offset	66.60 ± 1.34	63.65 ± 1.55	15.42 ± 0.49
CIFAR-100	FeNeC-E + offset	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	FeNeC-M (native)	59.11 ± 0.85	53.58 ± 1.27	22.91 ± 0.31
CIFAR-100	FeNeC-E (native)	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	Tip-Adapter sum (native)	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean (native)	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
Pets	Retained classifier	94.09 ± 2.37	91.77 ± 3.16	0.00 ± 0.00
Pets	Classifier removed (fixed gates)	77.17 ± 26.01	76.85 ± 23.25	16.92 ± 28.33
Pets	No classifier (recalibrated)	94.20 ± 2.05	92.57 ± 2.31	-0.11 ± 0.56
Pets	Anchored text + prototypes	94.35 ± 0.60	92.48 ± 1.10	-0.26 ± 1.82
Pets	Tip-Adapter sum + offset	93.38 ± 2.32	91.50 ± 2.78	0.71 ± 1.21
Pets	Tip-Adapter mean + offset	94.47 ± 2.38	92.88 ± 2.51	-0.37 ± 1.61
Pets	FeNeC-M + offset	80.34 ± 1.29	75.66 ± 1.72	13.76 ± 1.61
Pets	FeNeC-E + offset	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	FeNeC-M (native)	73.66 ± 4.73	66.74 ± 6.13	20.44 ± 2.69
Pets	FeNeC-E (native)	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	Tip-Adapter sum (native)	94.35 ± 1.08	92.72 ± 1.65	-0.25 ± 1.32
Pets	Tip-Adapter mean (native)	95.15 ± 1.28	93.74 ± 1.29	-1.06 ± 1.33
ImageNet-R	Retained classifier	87.23 ± 0.21	85.40 ± 0.36	0.00 ± 0.00
ImageNet-R	Classifier removed (fixed gates)	77.13 ± 6.75	74.64 ± 7.19	10.09 ± 6.55
ImageNet-R	No classifier (recalibrated)	85.01 ± 0.57	83.65 ± 0.70	2.22 ± 0.40
ImageNet-R	Anchored text + prototypes	86.36 ± 0.24	84.67 ± 0.39	0.87 ± 0.29
ImageNet-R	Tip-Adapter sum + offset	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean + offset	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35
ImageNet-R	FeNeC-M + offset	59.12 ± 0.54	55.79 ± 0.63	28.11 ± 0.46
ImageNet-R	FeNeC-E + offset	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	FeNeC-M (native)	54.57 ± 0.47	49.71 ± 0.32	32.65 ± 0.37
ImageNet-R	FeNeC-E (native)	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	Tip-Adapter sum (native)	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean (native)	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35

Appendix G.2. Repeated calibration and support budgets

The 81 configurations span three datasets, three original class partitions, $k \in \{1, 2, 5\}$ and three joint calibration/support trials. Trial zero reuses the original pseudo split. The other trials select $\lfloor |\mathcal{B}|/2 \rfloor$ pseudo-base classes with seeds $1701 + p$ and $2701 + p$. Pseudo classifiers select ridge regularization on pseudo-base validation. Support-index seeds $7, 8, 9$ select one- and two-shot subsets from each five-image reserve. At five shots, the support set is identical across trials: variation comes from the pseudo split. These are correlated trials of the same underlying class partitions. We average the orders and trials within each partition and report the mean and sample standard deviation of the three partition means.

Each configuration independently calibrates the retained, recalibrated-removal, dense-prototype and dynamic-reference readouts on the same base-only episode. Fixed-gate removal preserves the retained mixture's remaining parameters. Dynamic normalization uses the current text candidates and current novel prototypes as references, with the same learned branch and calibration grid as the retained table. Its own mixture is selected independently and stays fixed across arrivals. All retention targets are shared within each configuration. Feasibility counts refer to nine partition/trial combinations per support budget. When only some trials are feasible, the group accuracies average those trials within each partition before summarizing partitions; paired constrained comparisons use the intersection of feasible trial IDs.

Fixed references have zero restricted pre-offset candidate-ranking changes in all 648 append checks (81 configurations, two orders, four appends). The largest floating-point score differences are 1.40×10^{-4} on CIFAR, 5.36×10^{-6} on

Table S12: Calibration and support robustness. Average incremental and final accuracies use fixed gates. Paired gains are retained minus each independently calibrated comparator, after averaging two arrival orders and three trials within each partition. Values are percentages or percentage points, mean ± sample standard deviation across three partitions.

Data	Shots	Retained avg.	Δ recalibrated	Δ dense	Δ dynamic	Retained final
CIFAR-100	1	79.77 ± 0.61	0.55 ± 1.15	0.08 ± 0.72	0.10 ± 0.74	76.32 ± 1.22
CIFAR-100	2	80.20 ± 0.32	0.84 ± 0.67	-0.76 ± 0.52	0.55 ± 0.84	77.00 ± 0.85
CIFAR-100	5	80.57 ± 0.96	1.22 ± 1.18	-0.79 ± 1.08	0.62 ± 0.59	77.25 ± 1.09
Pets	1	93.94 ± 2.37	0.05 ± 0.67	-0.24 ± 0.48	-0.54 ± 0.11	91.60 ± 3.20
Pets	2	92.64 ± 1.82	-1.45 ± 1.19	-1.62 ± 1.15	-1.43 ± 1.09	90.09 ± 1.87
Pets	5	94.56 ± 1.74	0.78 ± 1.03	0.43 ± 0.80	0.47 ± 1.43	92.75 ± 2.00
ImageNet-R	1	86.16 ± 0.22	1.81 ± 0.33	1.01 ± 0.29	-0.07 ± 0.29	83.83 ± 0.14
ImageNet-R	2	86.58 ± 0.37	2.10 ± 0.28	0.62 ± 0.34	-0.07 ± 0.28	84.43 ± 0.67
ImageNet-R	5	87.16 ± 0.12	2.14 ± 0.37	0.54 ± 0.23	0.15 ± 0.19	85.36 ± 0.25

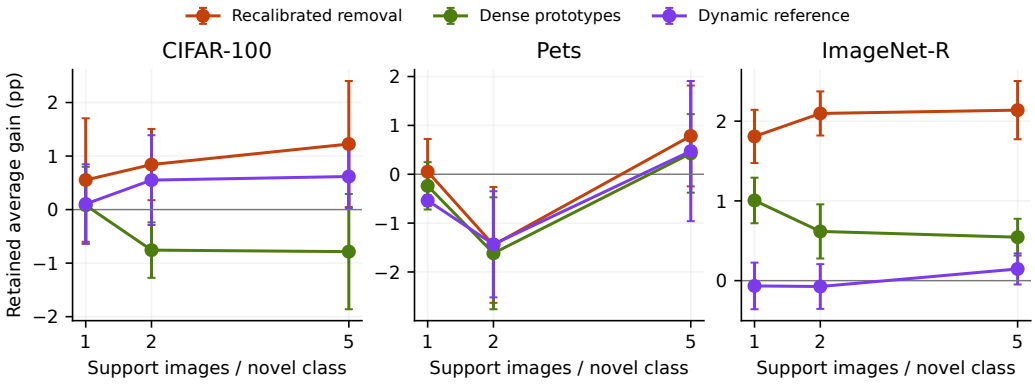


Figure S4: Paired average incremental gains under one-, two- and five-shot support. Error bars are partition-level sample standard deviations after averaging correlated calibration/support trials and arrival orders. The horizontal zero line identifies which readout has higher average accuracy in each setting.

Pets and 2.19×10^{-5} on ImageNet-R. Dynamic references produce maximum score changes of 2.40, 1.78 and 1.30, respectively, and up to 429, 206 and 118 restricted prediction changes per append over the full cached test query pool. These diagnostics compare the same previously active candidate IDs before adding the fixed group offset; they isolate changes induced by reference statistics. Recognition accuracy is reported separately in the fixed-gate and retention-policy tables.

Table S13: CIFAR-100: final-session outcomes in the repeated-calibration study at the 95% budget. Match uses base test labels descriptively; Val. uses base validation only and reports achieved test accuracy. Feasibility is counted over nine correlated partition/trial combinations.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	83.54 ± 0.81	73.03 ± 3.08	9/9	80.77 ± 2.34	76.30 ± 3.82	9/9
1	No classifier (recalibrated)	83.37 ± 0.57	59.17 ± 3.51	8/9	80.92 ± 1.45	72.28 ± 4.17	9/9
1	Anchored text + prototypes	83.37 ± 0.57	63.65 ± 3.03	8/9	80.38 ± 2.28	73.74 ± 5.16	9/9
1	Dynamic reference	83.54 ± 0.81	73.07 ± 3.81	9/9	81.10 ± 2.54	75.86 ± 4.56	9/9
2	Retained classifier	82.99 ± 0.46	73.84 ± 3.52	9/9	80.65 ± 1.93	76.40 ± 4.33	9/9
2	No classifier (recalibrated)	82.99 ± 0.46	61.20 ± 0.62	9/9	81.14 ± 1.01	71.44 ± 2.35	9/9
2	Anchored text + prototypes	82.99 ± 0.46	71.62 ± 6.74	9/9	80.81 ± 1.65	76.62 ± 4.51	9/9
2	Dynamic reference	82.99 ± 0.46	74.70 ± 3.56	9/9	80.37 ± 1.53	77.66 ± 2.64	9/9
5	Retained classifier	83.83 ± 0.76	75.71 ± 2.77	9/9	82.03 ± 1.38	77.69 ± 1.64	9/9
5	No classifier (recalibrated)	83.55 ± 0.29	62.19 ± 6.29	7/9	81.21 ± 1.18	73.83 ± 6.90	7/9
5	Anchored text + prototypes	83.83 ± 0.76	73.28 ± 3.14	9/9	81.95 ± 0.54	75.45 ± 1.14	8/9
5	Dynamic reference	83.83 ± 0.76	73.83 ± 0.74	9/9	82.44 ± 1.94	76.58 ± 2.18	9/9

Table S14: Oxford-IIIT Pets: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	92.58 ± 1.83	93.50 ± 0.89	9/9	88.07 ± 1.16	94.84 ± 0.89	9/9
1	No classifier (recalibrated)	92.58 ± 1.85	92.30 ± 2.36	9/9	88.56 ± 1.16	95.45 ± 0.70	6/9
1	Anchored text + prototypes	92.57 ± 1.84	93.03 ± 1.60	9/9	87.59 ± 1.34	94.59 ± 1.55	9/9
1	Dynamic reference	92.56 ± 1.82	93.74 ± 1.02	9/9	87.89 ± 1.50	95.47 ± 0.16	8/9
2	Retained classifier	92.28 ± 2.51	94.42 ± 1.14	9/9	88.05 ± 2.52	95.66 ± 0.49	9/9
2	No classifier (recalibrated)	92.27 ± 2.52	93.97 ± 0.96	9/9	87.04 ± 3.11	95.89 ± 0.27	7/9
2	Anchored text + prototypes	92.27 ± 2.50	94.33 ± 0.58	9/9	87.64 ± 2.55	95.51 ± 0.58	9/9
2	Dynamic reference	92.27 ± 2.50	94.62 ± 1.31	9/9	87.85 ± 2.40	95.77 ± 0.27	9/9
5	Retained classifier	92.40 ± 2.41	94.95 ± 1.39	9/9	88.31 ± 2.40	96.06 ± 0.47	9/9
5	No classifier (recalibrated)	92.40 ± 2.42	94.08 ± 0.92	9/9	87.39 ± 2.71	96.12 ± 0.41	7/9
5	Anchored text + prototypes	92.39 ± 2.39	94.79 ± 0.92	9/9	87.94 ± 2.68	95.99 ± 0.80	9/9
5	Dynamic reference	92.40 ± 2.40	94.84 ± 1.36	9/9	87.44 ± 2.34	96.12 ± 0.27	9/9

Table S15: ImageNet-R: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	86.24 ± 0.38	81.59 ± 0.50	9/9	85.73 ± 0.71	81.95 ± 0.67	9/9
1	No classifier (recalibrated)	86.25 ± 0.37	70.69 ± 10.24	9/9	86.13 ± 0.37	69.56 ± 13.12	9/9
1	Anchored text + prototypes	86.24 ± 0.36	78.57 ± 0.59	9/9	86.14 ± 0.54	78.97 ± 0.47	9/9
1	Dynamic reference	86.25 ± 0.36	81.78 ± 0.76	9/9	85.87 ± 0.61	82.11 ± 1.07	9/9
2	Retained classifier	86.38 ± 0.34	82.67 ± 1.21	9/9	86.10 ± 0.63	82.86 ± 1.30	9/9
2	No classifier (recalibrated)	86.38 ± 0.33	69.12 ± 14.06	9/9	86.27 ± 0.38	58.73 ± 33.47	9/9
2	Anchored text + prototypes	86.39 ± 0.34	81.53 ± 2.15	9/9	86.40 ± 0.40	81.50 ± 1.68	9/9
2	Dynamic reference	86.38 ± 0.33	82.80 ± 0.65	9/9	86.04 ± 0.67	83.02 ± 0.76	9/9
5	Retained classifier	86.35 ± 0.27	84.48 ± 0.35	9/9	86.08 ± 0.44	84.69 ± 0.46	9/9
5	No classifier (recalibrated)	86.34 ± 0.27	73.89 ± 9.72	9/9	86.36 ± 0.21	63.98 ± 26.37	9/9
5	Anchored text + prototypes	86.34 ± 0.27	83.45 ± 1.62	9/9	86.23 ± 0.52	83.65 ± 1.19	9/9
5	Dynamic reference	86.35 ± 0.29	84.42 ± 0.56	9/9	86.21 ± 0.47	84.50 ± 0.69	9/9

Appendix G.3. Readout cost and restart verification

Table S16 measures 512 cached queries on the RTX 4070 Laptop in float32. Each readout receives five warm-ups and 20 synchronized timing repetitions. Static cache values are constructed before timing, query chunks contain 256 images for retrieval, and unused score branches are skipped. An independently checked implementation verifies score agreement within 10^{-3} for every optimized readout. Persistent tensor bytes count text anchors, learned weights and references/prototypes for the tables, or text anchors, cache keys and the stored dense one-hot/class-mean value matrix for retrieval. These are implementation-level readout costs, excluding the common image encoder, query features and temporary workspaces. They do not represent complete model memory or end-to-end image inference time.

Every calibration and arrival-session result is written atomically. A live pause/resume check preserved a completed session's SHA-256 hash and modification time, then completed all 90 configuration units. The numerical verifier reconstructs group predictions, macro weights, selected thresholds and test metrics from archived sufficient statistics, including the base-validation

Table S16: Measured readout latency and persistent tensor storage for a final-session 512-query batch on the same laptop GPU. MiB denotes 2^{20} bytes. Cache keys number 4,937 on CIFAR, 553 on Pets and 3,830 on ImageNet-R in partition zero.

Data	Readout	Median (ms)	P90 (ms)	Stored tensors (MiB)
CIFAR-100	Retained classifier	0.715	0.763	0.586
CIFAR-100	Dense text + prototypes	0.515	0.542	0.469
CIFAR-100	Tip-Adapter sum	0.838	0.862	16.205
CIFAR-100	Tip-Adapter mean	0.841	0.857	16.205
Pets	Retained classifier	0.880	1.280	0.220
Pets	Dense text + prototypes	0.586	0.680	0.176
Pets	Tip-Adapter sum	0.564	0.673	1.771
Pets	Tip-Adapter mean	0.569	0.732	1.771
ImageNet-R	Retained classifier	0.864	0.918	1.465
ImageNet-R	Dense text + prototypes	0.631	0.790	1.172
ImageNet-R	Tip-Adapter sum	0.759	0.873	14.729
ImageNet-R	Tip-Adapter mean	0.732	0.898	14.729

selection arrays. The trial-zero five-shot table results agree exactly with the prior three-dataset records. Source, protocol and cache hashes, all outcomes and reproduction commands accompany the release.

Appendix H. Validation-query budget for admission thresholds

This experiment varies the old-class query labels used to select the admission threshold, conditional on the original five-shot classifier and fusion calibration. All fitted parameters, cache settings and branch mixtures stay fixed. The evaluation uses the original trial-zero score archives, class partitions and arrival order zero on each dataset. No image encoder or new classifier is fitted.

For each base class with n_c validation images, the nominal fraction f keeps $\max(1, \lfloor f n_c \rfloor)$ images. Fractions $f \in \{0.25, 0.5\}$ use three predeclared subset seeds, with nested class-wise permutations; the full budget uses all images once. Actual query counts are reported because rounding and the one-query

minimum change the realized fractions. Macro weights are recomputed from the selected class counts. The retained readout's closed-base accuracy on that subset defines a shared target $rA^{\text{closed}}_{B,\text{ref,subset}}$ for all six readouts, with $r \in \{0.95, 0.99\}$. The largest feasible offset uses only the selected base-validation margins and labels.

After selection, test queries measure both group accuracies. A test target is met when $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$; this criterion is evaluated after the offset is fixed and never enters threshold selection. Table S17 separates validation feasibility from test target attainment. Subset repeats are averaged within each class partition before the mean and sample standard deviation over available partition means. The nine reduced-budget cases share three underlying partitions and are not independent repetitions. Small per-class counts quantize the selected threshold, so different target ratios can yield identical predictions and increasing the validation fraction need not change test target attainment monotonically. These empirical policies do not supply a finite-sample guarantee on test retention.

The full-budget policies reproduce the original validation-selected outcomes exactly. Saved predictions reconstruct every offset, macro accuracy and attainment flag across 315 session units and 3,780 metric records. The complete budgets, query fractions and test-retention ratios are in `validation_budget_summary.csv`. This experiment varies threshold-selection queries while holding the initial full-data fusion calibration fixed.

Table S17: Retained readout under different validation-query budgets after the final arrival. Accuracies are percentages, mean ± partition-level sample standard deviation after averaging subset repeats. Queries gives the actual old-class validation count range. Val. feasible counts partition/subset combinations meeting the validation constraint; Test met counts feasible combinations meeting the separate test criterion. The full budget uses one subset per partition.

Data	Nominal budget	Queries	Target	Base test	Novel test	Val. feasible	Test met
CIFAR-100	25%	40	95%	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	25%	40	99%	85.21 ± 2.64	66.99 ± 9.18	9/9	4/9
CIFAR-100	50%	68	95%	81.94 ± 3.23	76.92 ± 3.72	9/9	3/9
CIFAR-100	50%	68	99%	85.85 ± 2.61	64.30 ± 10.71	9/9	5/9
CIFAR-100	100%	161–163	95%	82.99 ± 2.29	75.71 ± 3.14	3/3	1/3
CIFAR-100	100%	161–163	99%	86.48 ± 1.65	62.83 ± 10.11	3/3	2/3
Pets	25%	15	95%	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	25%	15	99%	91.77 ± 2.39	92.42 ± 2.64	9/9	4/9
Pets	50%	25	95%	92.02 ± 0.95	94.75 ± 1.23	9/9	5/9
Pets	50%	25	99%	95.05 ± 0.48	90.49 ± 5.09	9/9	5/9
Pets	100%	60	95%	90.29 ± 2.90	95.68 ± 0.83	3/3	0/3
Pets	100%	60	99%	96.43 ± 2.15	88.67 ± 7.89	3/3	2/3
ImageNet-R	25%	696–731	95%	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	25%	696–731	99%	89.79 ± 0.16	74.81 ± 2.33	9/9	5/9
ImageNet-R	50%	1444–1519	95%	85.93 ± 0.65	84.87 ± 0.63	9/9	1/9
ImageNet-R	50%	1444–1519	99%	89.90 ± 0.33	74.24 ± 2.68	9/9	5/9
ImageNet-R	100%	2938–3087	95%	86.01 ± 0.59	84.73 ± 0.71	3/3	0/3
ImageNet-R	100%	2938–3087	99%	89.94 ± 0.25	74.48 ± 1.35	3/3	2/3

Table S18: All readouts at the nominal 25% validation-query budget and 95% target after the final arrival. All classifiers, mixtures and cache settings are held fixed from the original five-shot calibration. Accuracy aggregation and feasibility/attainment counts follow Table S17. Missing metrics correspond to zero feasible subsets.

Data	Readout	Base test	Novel test	Val. feasible	Test met
CIFAR-100	Retained classifier	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	Fixed-gate removal	79.39 ± 2.08	74.84 ± 0.15	5/9	1/5
CIFAR-100	Recalibrated removal	79.48 ± 2.19	73.22 ± 2.45	5/9	1/5
CIFAR-100	Dense text + prototypes	81.30 ± 2.08	77.12 ± 2.64	7/9	0/7
CIFAR-100	Tip-Adapter sum	79.32 ± 2.30	71.40 ± 7.86	6/9	1/6
CIFAR-100	Tip-Adapter mean	79.51 ± 3.58	77.12 ± 5.00	8/9	1/8
Pets	Retained classifier	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	Fixed-gate removal	88.87 ± 3.95	92.77 ± 5.91	6/9	1/6
Pets	Recalibrated removal	87.68 ± 3.95	89.64 ± 11.35	6/9	1/6
Pets	Dense text + prototypes	88.50 ± 6.27	93.41 ± 3.82	7/9	2/7
Pets	Tip-Adapter sum	89.86 ± 3.58	89.59 ± 5.68	7/9	2/7
Pets	Tip-Adapter mean	90.25 ± 1.80	91.90 ± 3.31	7/9	2/7
ImageNet-R	Retained classifier	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	Fixed-gate removal	85.66 ± 0.86	79.23 ± 1.90	7/9	3/7
ImageNet-R	Recalibrated removal	85.74 ± 0.90	79.08 ± 1.74	7/9	3/7
ImageNet-R	Dense text + prototypes	86.39 ± 0.50	83.05 ± 0.22	9/9	4/9
ImageNet-R	Tip-Adapter sum	85.47 ± 0.89	77.71 ± 1.44	8/9	4/8
ImageNet-R	Tip-Adapter mean	86.49 ± 0.54	81.22 ± 0.63	9/9	6/9

Appendix I. Ten-partition replication, encoder transfer and simple fusion

The expanded experiment fixes class-partition seeds $0, \ldots, 9$ on each of the three existing image pools. Assignments and pseudo splits $0, 1, 2$ are retained exactly, and the original L/14 results reconstruct with maximum metric difference zero. New seeds use the same frequency-stratified allocation, base/novel counts, five-shot reserve, pseudo-episode selection and two arrival-order rules. These are ten distinct class assignments on shared images, not ten independent datasets. Primary fixed-gate comparisons average both orders within a partition. The conditional paired percentile bootstrap uses 20,000 resamples of the ten paired partition values with seed 2026. Constrained differences use identical jointly feasible partition/order/session cells. JSON records retain the partition values and available sets, and CSV tables report every budget.

Appendix I.1. Second encoder and access controls

OpenAI CLIP ViT-B/32 uses cached revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268` with 512-dimensional projected features; L/14 uses 768 dimensions. Both consume the exact same fitting, validation, support and test images, dataset-specific preprocessing and eight text templates. Text anchors and ridge columns are recomputed for the selected encoder; no task-adapted features are used. The two encoders have separate closed-base references and hence separate absolute targets. Config, weights, tokenizer and pool hashes are recorded. Batch-16 feature shards support atomic restart with contiguous-index guards, and eight fresh test images are re-encoded per dataset to check each completed B/32 cache.

Appendix I.2. Raw and fixed-temperature controls

Raw fusion uses q_T, q_L and q_P directly in the same availability pattern as the retained table. Fixed-temperature fusion uses q_e/σ_e, where

$$\sigma_e = \max\left(10^{-6}, \frac{1}{|\mathcal{V}_B|}\sum_{x\in\mathcal{V}_B} \mathrm{Std}_{r\in\mathcal{R}_e}\, q_e(x,r)\right).$$

For the pseudo episode, $\mathcal{V}_B$ contains base-validation queries and $\mathcal{R}_e$ contains pseudo-base reference columns. Full-model scales use full base columns and base validation before arrivals. The three scalars remain fixed across sessions. Coefficient and offset grids are identical to the main retained model, with the same harmonic pseudo-episode objective. Neither control requires current-vocabulary statistics; both can preserve old scores. Their performance tests the empirical value of the per-query normalization, conditional on this specified calibration grid.

Appendix I.3. Full matched and validation-policy results

The main validation table includes retained, recalibrated-removal, dense-prototype, class-mean cache and simple-fusion alternatives. Summed cache and fixed-gate removal remain in the complete records. The following tables give both encoders' matched-test frontiers and the B/32 validation policy at the 95% budget. Every feasible set and test-attainment count refers to the final active vocabulary; order zero counts each partition once. Because means may cover different feasible sets, an unpaired difference of available means is not reported as a paired gain. Complete 80/85/90/95/99% outcomes are retained in `revision_summary.csv`; per-partition raw predictions supply both test and validation sufficient statistics.

Table S19: Complete eight-readout ViT-L/14 validation-selected 95% policy. Accuracies are achieved test percentages, mean ± sample standard deviation over feasible partitions. Feasible and Test met have the same meanings as the main validation-policy table.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Fixed-gate removal	83.05 ± 1.64	65.32 ± 12.41	6/10	1/6
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
CIFAR-100	Tip-Adapter sum	83.21 ± 1.71	65.36 ± 12.68	5/10	0/5
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Fixed-gate removal	90.43 ± 3.16	93.81 ± 3.11	8/10	3/8
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
Pets	Tip-Adapter sum	89.18 ± 1.73	91.61 ± 6.00	7/10	1/7
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Fixed-gate removal	86.17 ± 0.65	66.89 ± 22.80	10/10	3/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10
ImageNet-R	Tip-Adapter sum	86.06 ± 0.78	74.52 ± 3.69	10/10	2/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10

Table S20: ViT-L/14 matched-test 95% retention frontier across ten class partitions. Values are percentages, mean ± sample standard deviation over feasible partitions after averaging orders. Test met equals feasibility by construction in this descriptive sweep.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	83.91 ± 1.42	78.02 ± 2.27	10/10	10/10
CIFAR-100	Fixed-gate removal	83.25 ± 1.17	61.68 ± 9.35	6/10	6/6
CIFAR-100	Recalibrated removal	83.25 ± 1.17	63.78 ± 6.52	6/10	6/6
CIFAR-100	Dense text + prototypes	83.91 ± 1.42	76.18 ± 2.56	10/10	10/10
CIFAR-100	Raw-score fusion	83.25 ± 1.17	62.37 ± 6.38	6/10	6/6
CIFAR-100	Fixed temperatures	83.64 ± 1.21	74.53 ± 2.70	9/10	9/9
CIFAR-100	Tip-Adapter sum	83.16 ± 1.28	61.83 ± 4.92	5/10	5/5
CIFAR-100	Tip-Adapter mean	83.91 ± 1.42	65.20 ± 11.69	10/10	10/10
Pets	Retained references	91.56 ± 1.87	93.30 ± 3.74	10/10	10/10
Pets	Fixed-gate removal	91.57 ± 1.88	93.78 ± 2.17	10/10	10/10
Pets	Recalibrated removal	91.56 ± 1.88	93.32 ± 2.10	10/10	10/10
Pets	Dense text + prototypes	91.56 ± 1.87	93.79 ± 2.78	10/10	10/10
Pets	Raw-score fusion	91.57 ± 1.88	92.39 ± 2.75	10/10	10/10
Pets	Fixed temperatures	91.56 ± 1.87	93.64 ± 2.48	10/10	10/10
Pets	Tip-Adapter sum	91.51 ± 1.98	87.78 ± 9.98	9/10	9/9
Pets	Tip-Adapter mean	91.57 ± 1.87	92.02 ± 4.15	10/10	10/10
ImageNet-R	Retained references	86.42 ± 0.53	84.31 ± 1.01	10/10	10/10
ImageNet-R	Fixed-gate removal	86.41 ± 0.56	73.82 ± 6.70	9/10	9/9
ImageNet-R	Recalibrated removal	86.42 ± 0.56	73.50 ± 6.79	9/10	9/9
ImageNet-R	Dense text + prototypes	86.42 ± 0.53	83.02 ± 1.40	10/10	10/10
ImageNet-R	Raw-score fusion	86.42 ± 0.56	70.85 ± 8.03	9/10	9/9
ImageNet-R	Fixed temperatures	86.43 ± 0.53	82.49 ± 1.11	10/10	10/10
ImageNet-R	Tip-Adapter sum	86.43 ± 0.56	73.16 ± 5.66	9/10	9/9
ImageNet-R	Tip-Adapter mean	86.43 ± 0.52	80.91 ± 1.48	10/10	10/10

Table S21: ViT-B/32 matched-test 95% frontier under the same access, aggregation and feasibility reporting as Table S20. Absolute targets are encoder-specific.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	76.16 ± 1.99	59.64 ± 2.10	10/10	10/10
CIFAR-100	Fixed-gate removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Recalibrated removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Dense text + prototypes	76.15 ± 2.22	51.78 ± 8.42	8/10	8/8
CIFAR-100	Raw-score fusion	75.12 ± 1.63	46.30 ± 9.93	5/10	5/5
CIFAR-100	Fixed temperatures	75.71 ± 1.46	54.14 ± 4.58	9/10	9/9
CIFAR-100	Tip-Adapter sum	74.08 ± 1.05	52.81 ± 5.29	3/10	3/3
CIFAR-100	Tip-Adapter mean	76.16 ± 2.11	46.50 ± 11.77	9/10	9/9
Pets	Retained references	86.92 ± 2.17	76.54 ± 6.71	10/10	10/10
Pets	Fixed-gate removal	86.45 ± 2.27	77.62 ± 6.10	5/10	5/5
Pets	Recalibrated removal	86.46 ± 2.29	78.61 ± 5.07	5/10	5/5
Pets	Dense text + prototypes	86.84 ± 2.33	71.48 ± 15.80	8/10	8/8
Pets	Raw-score fusion	87.12 ± 2.19	67.25 ± 11.41	9/10	9/9
Pets	Fixed temperatures	86.93 ± 2.16	74.97 ± 6.20	10/10	10/10
Pets	Tip-Adapter sum	86.83 ± 2.22	70.08 ± 9.72	6/10	6/6
Pets	Tip-Adapter mean	87.28 ± 2.29	68.64 ± 21.10	8/10	8/8
ImageNet-R	Retained references	70.38 ± 0.87	56.36 ± 1.11	10/10	10/10
ImageNet-R	Fixed-gate removal	69.72	31.36	1/10	1/1
ImageNet-R	Recalibrated removal	69.71	30.54	1/10	1/1
ImageNet-R	Dense text + prototypes	70.37 ± 0.88	48.30 ± 3.45	10/10	10/10
ImageNet-R	Raw-score fusion	69.71	28.06	1/10	1/1
ImageNet-R	Fixed temperatures	70.38 ± 0.87	53.72 ± 1.35	10/10	10/10
ImageNet-R	Tip-Adapter sum	69.67 ± 0.03	35.63 ± 8.58	2/10	2/2
ImageNet-R	Tip-Adapter mean	70.45 ± 0.92	41.88 ± 7.58	8/10	8/8

Table S22: ViT-B/32 validation-selected 95% policy. Values are achieved base/novel test percentages over validation-feasible partitions; Test met counts those meeting the independent test-retention criterion. Validation feasibility does not guarantee test target attainment.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	75.98 ± 2.43	59.04 ± 3.66	10/10	5/10
CIFAR-100	Fixed-gate removal	73.46 ± 1.73	52.54 ± 10.71	7/10	1/7
CIFAR-100	Recalibrated removal	73.40 ± 1.79	54.20 ± 6.94	7/10	1/7
CIFAR-100	Dense text + prototypes	74.30 ± 2.85	55.73 ± 8.03	7/10	2/7
CIFAR-100	Raw-score fusion	73.82 ± 2.67	49.47 ± 12.37	8/10	2/8
CIFAR-100	Fixed temperatures	73.90 ± 2.39	56.16 ± 8.19	9/10	1/9
CIFAR-100	Tip-Adapter sum	73.55 ± 2.53	49.25 ± 12.78	7/10	1/7
CIFAR-100	Tip-Adapter mean	73.54 ± 2.88	57.47 ± 8.46	7/10	0/7
Pets	Retained references	88.16 ± 3.34	72.75 ± 8.65	10/10	8/10
Pets	Fixed-gate removal	86.17 ± 2.74	76.30 ± 6.46	5/10	3/5
Pets	Recalibrated removal	86.24 ± 2.14	77.70 ± 5.37	5/10	2/5
Pets	Dense text + prototypes	86.89 ± 3.09	70.27 ± 15.78	7/10	3/7
Pets	Raw-score fusion	86.93 ± 5.36	63.79 ± 15.01	9/10	5/9
Pets	Fixed temperatures	85.53 ± 5.27	74.57 ± 7.40	10/10	4/10
Pets	Tip-Adapter sum	86.87 ± 2.42	73.74 ± 4.16	4/10	2/4
Pets	Tip-Adapter mean	84.97 ± 3.13	81.94 ± 3.51	5/10	1/5
ImageNet-R	Retained references	70.18 ± 0.96	56.80 ± 1.33	10/10	3/10
ImageNet-R	Fixed-gate removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Recalibrated removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Dense text + prototypes	70.43 ± 1.50	46.95 ± 4.97	10/10	6/10
ImageNet-R	Raw-score fusion	68.16 ± 0.84	21.79 ± 6.87	2/10	0/2
ImageNet-R	Fixed temperatures	70.46 ± 1.26	53.38 ± 1.43	10/10	6/10
ImageNet-R	Tip-Adapter sum	68.89 ± 0.79	38.12 ± 2.27	3/10	0/3
ImageNet-R	Tip-Adapter mean	69.99 ± 1.61	46.39 ± 3.62	9/10	4/9

Appendix J. Independent protection-risk calibration

Appendix J.1. Disjoint access and refitting

For each dataset, original fitting and validation features are concatenated in that order. A generator with seed 662026 shuffles each class, reserving $\max(1, \lfloor 0.4N_c \rfloor)$ calibration and $\max(1, \lfloor 0.2N_c \rfloor)$ tuning observations; the remainder fits the readouts. Every class retains at least two fitting images. Identical indices are used by both encoders and every method. The global fitting/tuning/calibration sizes are 4,437/2,139/4,318 on CIFAR, 556/248/510 on Pets, and 5,096/2,390/4,863 on ImageNet-R. Each true partition uses only its base classes in these pools. Support and test observations remain outside all three. All ridge parameters, pseudo-episode mixtures, branch temperatures and semantic-cache hyperparameters are recomputed using fitting and tuning data exclusively.

Only frozen features are reused; previously tuned score models are not treated as independent of their old validation labels. The new calibration labels enter only threshold certification or the explicitly compared empirical threshold controls. The independent-pool experiment is a benchmark revision on previously studied test data, not a newly untouched final evaluation. The statistical analysis assumes that, conditional on each base class and the fitted models, calibration observations are independent draws from the class-conditional distribution to which the population statement applies. Dataset shift and adaptive support arrivals depending on calibration labels are outside that statement.

Appendix J.2. Weighted protection bound

Let B be the number of base classes and n_c their calibration counts. Assign each observation of class c weight $w_i = 1/(Bn_c)$, so $\sum_i w_i = 1$, and let $b = \max_i w_i$. For any fixed bounded losses $L_i \in [0, 1]$, write $q = \sum_i w_i L_i$ and $\mu = \sum_i w_i \mathbb{E}L_i$. Conditional on the score models and the independently selected threshold family, independence gives, for $t < 0$,

$$\log \mathbb{E}e^{tq} \le \sum_i \log(1 - \mu_i + \mu_i e^{tw_i}) \tag{J.1}$$

$$\le b^{-1} \sum_i w_i \log(1 - \mu_i + \mu_i e^{tb}) \tag{J.2}$$

$$\le b^{-1} \log(1 - \mu + \mu e^{tb}), \tag{J.3}$$

where $\mu_i = \mathbb{E}L_i$. The first inequality uses boundedness and the chord of the exponential; the second uses convexity of each log-moment-generating function in w_i and its zero value at zero; the third uses concavity in μ_i. Chernoff optimization yields

$$\Pr(q \le x) \le \exp[-\operatorname{kl}(x, \mu)/b], \qquad x < \mu, \tag{J.4}$$

where kl is Bernoulli relative entropy. Thus, for the null $\mu \ge \alpha$, the value

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \ge \alpha \end{cases} \tag{J.5}$$

is a valid lower-tail p-value. Unlike an ordinary pooled binomial calculation, this bound retains the unequal macro weights. Its dependence on the largest weight makes sparse class-conditional calibration sets conservative.

Apply the bound to the protection loss of main Eq. (15). Let $A_{\mathrm{ref}} = \Pr(R = 1)$, $D = \Pr(R = 1, C_d = 0)$ and $G = \Pr(R = 0, C_d = 1)$. Then

$$\mathbb{E}L_d = D + (1 - r)(1 - A_{\mathrm{ref}}), \qquad A_B(d) = A_{\mathrm{ref}} - D + G. \tag{J.6}$$

Consequently $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{\mathrm{ref}}$. The reference uncertainty is included in the paired loss rather than replaced by a point estimate. Retained base scores have $G = 0$; other readouts may have $G > 0$, so protection is sufficient and can be stricter than accuracy-ratio control for replacement models.

Appendix J.3. Threshold search and trajectory confidence

For each method and arrival, the independently fixed finite threshold family consists of unique quantiles at 0,1,. . . ,100% of base tuning margins, in increasing order of novel admission. Following fixed-sequence testing [21], test this family in order at level $\delta/5$, stop at the first failure, and return the last accepted threshold. A false certification requires rejecting the first true null in the fixed order, whose probability is at most $\delta/5$; independence between these tests is unnecessary. A union bound over the five prespecified arrivals gives error probability at most δ for each method/order/partition. The same calibration pool may be reused because the score models, threshold families and support arrivals do not adapt to its labels. No simultaneous confidence claim across methods, orders, partitions, datasets or separately reported δ values is made.

If the accepted finite set is empty, the policy returns the original reference classifier and admits no novel class. This fallback structurally preserves its original accuracy and has zero novel accuracy. It is reported separately from

useful certified expansion. The confidence levels 90,95,99% and empirical target fractions 95,97,99% were specified before the new experiment; the primary setting uses 95% retention and 95% confidence. Their complete results, worst shortfalls and five-session trajectories are saved in `independen t_risk_summary.json` and `independent_risk_summary.csv`.

Table S23: Complete ViT-L/14 independent-calibration comparison, with meanings and fallback accounting as in main Table 2. All operational means include every partition; certification counts identify finite accepted thresholds.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	7/10	76.28	0/10	10/10	88.29	0.00
CIFAR-100	Recalibrated removal	6/10	35.32	0/10	10/10	88.29	0.00
CIFAR-100	Dense prototypes	6/10	71.92	0/10	10/10	88.29	0.00
CIFAR-100	Raw fusion	5/10	34.46	0/10	10/10	88.29	0.00
CIFAR-100	Fixed temperature	8/10	71.36	0/10	10/10	88.29	0.00
CIFAR-100	Class-mean cache	4/10	65.73	0/10	10/10	88.29	0.00
Pets	Retained	1/10	93.07	0/10	10/10	96.05	0.00
Pets	Recalibrated removal	4/10	93.29	0/10	10/10	96.05	0.00
Pets	Dense prototypes	0/10	95.08	0/10	10/10	96.05	0.00
Pets	Raw fusion	4/10	86.90	0/10	10/10	96.05	0.00
Pets	Fixed temperature	3/10	91.87	0/10	10/10	96.05	0.00
Pets	Class-mean cache	1/10	92.50	0/10	10/10	96.05	0.00
ImageNet-R	Retained	7/10	83.45	10/10	10/10	89.73	77.64
ImageNet-R	Recalibrated removal	4/10	57.36	0/10	10/10	91.54	0.00
ImageNet-R	Dense prototypes	6/10	82.47	5/10	10/10	91.04	24.12
ImageNet-R	Raw fusion	3/10	64.80	4/10	10/10	91.08	26.42
ImageNet-R	Fixed temperature	5/10	81.28	10/10	10/10	89.76	65.69
ImageNet-R	Class-mean cache	4/10	79.37	2/10	10/10	91.18	6.20

Table S24: Complete ViT-B/32 independent-calibration comparison. Infeasible certification triggers the shared base-only reference fallback and zero novel accuracy.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	1/10	60.64	0/10	10/10	79.08	0.00
CIFAR-100	Recalibrated removal	2/10	47.79	0/10	10/10	79.08	0.00
CIFAR-100	Dense prototypes	2/10	52.74	0/10	10/10	79.08	0.00
CIFAR-100	Raw fusion	2/10	50.33	0/10	10/10	79.08	0.00
CIFAR-100	Fixed temperature	5/10	55.06	0/10	10/10	79.08	0.00
CIFAR-100	Class-mean cache	2/10	57.79	0/10	10/10	79.08	0.00
Pets	Retained	1/10	75.10	0/10	10/10	90.24	0.00
Pets	Recalibrated removal	10/10	28.49	0/10	10/10	90.24	0.00
Pets	Dense prototypes	4/10	54.67	0/10	10/10	90.24	0.00
Pets	Raw fusion	6/10	53.57	0/10	10/10	90.24	0.00
Pets	Fixed temperature	5/10	64.90	0/10	10/10	90.24	0.00
Pets	Class-mean cache	7/10	42.00	0/10	10/10	90.24	0.00
ImageNet-R	Retained	2/10	56.51	10/10	10/10	73.66	45.30
ImageNet-R	Recalibrated removal	9/10	3.58	0/10	10/10	75.01	0.00
ImageNet-R	Dense prototypes	4/10	43.48	0/10	10/10	75.01	0.00
ImageNet-R	Raw fusion	9/10	3.75	0/10	10/10	75.01	0.00
ImageNet-R	Fixed temperature	2/10	52.45	7/10	10/10	74.24	27.28
ImageNet-R	Class-mean cache	5/10	33.38	0/10	10/10	75.01	0.00

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